# Testing Inheritance Patterns

Biology 2e chapter 12 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

## Before predicting

Each kernel develops from a fertilization event. Use documented parent genotypes and a defined visible trait before predicting offspring ratios. Sample size and chance affect observed counts.

## Investigation question

How closely do observed kernel traits match an inheritance prediction?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

## How to use the virtual lab

The model represents a documented single-locus Pp x Pp cross: P is dominant for purple pigment in this simplified trait. Before counting, predict the two phenotype proportions. Choose the assigned ear, mark one of its three 48-kernel regions, and count the visible kernels using separate purple and yellow tally counters. Optional pencil marks help track counted kernels; marks do not enter a phenotype for you.

Select the assigned ear, then mark a 48-kernel sampling region before its colors are revealed. Use separate purple and yellow tally counters. Tapping a kernel only adds or removes a pencil mark; it does not tally automatically. Check the total and review your counts. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: kernels

| Condition / sample | Repeat | Source | Purple | Yellow |
|---|---|---|---|---|
| Ear A | 1 | | | |
| Ear B | 1 | | | |
| Ear C | 1 | | | |

## Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

| Item / band / cell | Measurement or classification | Reasoning / uncertainty |
|---|---|---|
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For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Compare observed and expected counts under a 3:1 prediction.

Calculate chi-square: sum (observed - expected)^2 / expected. What does that statistic measure?

Would a small deviation alone disprove Mendelian segregation?

## Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Generated kernel populations and original artwork, not photographs or supplier data. Real corn pigment genetics can involve more than one locus; apply the prediction only to a documented suitable cross.